

1. NAME OF THE MEDICINAL PRODUCT

Gefitinib Sandoz Film Coated Tablet 250mg

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each tablet contains 250 mg of gefitinib.

Excipient with known effect: Each tablet contains 163.5 mg of lactose (as monohydrate).

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Film-coated tablet.

Brown, round, biconvex, film-coated tablets impressed with "250" on one side and plain on the other side

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Gefitinib Sandoz is indicated for the treatment of adult patients with locally advanced or metastatic non-small cell lung cancer (NSCLC) with activating mutations of EGFR-TK.

4.2 Posology and method of administration

The recommended dose of Gefitinib Sandoz is one 250 mg tablet once a day, taken with or without food. If a dose of Gefitinib Sandoz is missed, it should be taken as soon as the patient remembers. If it is less than 12 hours to the next dose, the patient should not take the missed dose. Patients should not take a double dose (two doses at the same time) to make up for a forgotten dose.

Where dosing of whole tablets is not possible, such as patients who are only able to swallow liquids, tablets may be administered as a dispersion in water. The tablet should be dropped into half a glass of drinking water (non-carbonated), without crushing, and the glass stirred until the tablet has dispersed (approximately 15 minutes) and the contents subsequently drunk immediately. The glass should be rinsed with a further half glass of water and the contents drunk. The liquid can also be administered via a nasogastric tube.

Gefitinib Sandoz is not recommended for use in children or adolescents as safety and effectiveness in these patient populations has not been studied.

No dosage adjustment is required on the basis of patient age, body weight, gender, ethnicity or renal function or in patients with moderate to severe hepatic impairment due to liver metastases.

Dosage adjustment: Patients with poorly tolerated diarrhoea or skin adverse drug reactions may be successfully managed by providing a brief (up to 14 days) therapy interruption followed by reinstatement of the 250 mg dose.

4.3 Contraindications

Hypersensitivity to the active substance or to any of the excipients.

4.4 Special warnings and precautions for use

When considering the use of gefitinib as a treatment for locally advanced or metastatic NSCLC, it is important that EGFR mutation assessment of the tumour tissue is attempted for all patients. If a tumour sample is not evaluable, then circulating tumour DNA (ctDNA) obtained from a blood (plasma) sample may be used.

Only robust, reliable and sensitive test(s) with demonstrated utility for the determination of EGFR mutation status of tumours or ctDNA should be used to avoid false negative or false positive determinations.

Interstitial lung disease (ILD)

Interstitial lung disease (ILD) which may be acute in onset has been observed in patients receiving gefitinib, and some cases have been fatal. If patients experience worsening of respiratory symptoms such as dyspnoea, cough and fever,

gefitinib should be interrupted and the patient should be promptly investigated. If ILD is confirmed, gefitinib should be discontinued and the patient treated appropriately.

Hepatotoxicity and liver impairment

Liver function test abnormalities (including increases in alanine aminotransferase, aspartate aminotransferase, bilirubin) have been observed, uncommonly presenting as hepatitis.

There have been isolated reports of hepatic failure which in some cases led to fatal outcomes. Therefore, periodic liver function testing is recommended. Gefitinib should be used cautiously in the presence of mild to moderate changes in liver function. Discontinuation should be considered if changes are severe.

Impaired liver function due to cirrhosis has been shown to lead to increased plasma concentrations of Gefitinib.

Interactions with other medicinal products

CYP3A4 inducers may increase metabolism of gefitinib and decrease gefitinib plasma concentrations.

Therefore, concomitant administration of CYP3A4 inducers (e.g. phenytoin, carbamazepine, rifampicin, barbiturates or herbal preparations containing St John's wort/*Hypericum perforatum*) may reduce efficacy of the treatment and should be avoided.

In individual patients with CYP2D6 poor metaboliser genotype, treatment with a potent CYP3A4 inhibitor might lead to increased plasma levels of gefitinib. At initiation of treatment with a CYP3A4 inhibitor, patients should be closely monitored for gefitinib adverse reactions.

International normalised ratio (INR) elevations and/or bleeding events have been reported in some patients taking warfarin together with gefitinib. Patients taking warfarin and gefitinib concomitantly should be monitored regularly for changes in prothrombin time (PT) or INR.

Medicinal products that cause significant sustained elevation in gastric pH, such as proton-pump inhibitors and h2-antagonists may reduce bioavailability and plasma concentrations of gefitinib and, therefore, may reduce efficacy. Antacids if taken regularly close in time to administration of gefitinib may have a similar effect.

Further precautions for use

Patients should be advised to seek medical advice immediately if they experience severe or persistent diarrhoea, nausea, vomiting or anorexia as these may indirectly lead to dehydration. These symptoms should be managed as clinically indicated.

Patients presenting with signs and symptoms suggestive of keratitis such as acute or worsening: eye inflammation, lacrimation, light sensitivity, blurred vision, eye pain and/or red eye should be referred promptly to an ophthalmology specialist.

If a diagnosis of ulcerative keratitis is confirmed, treatment with gefitinib should be interrupted, and if symptoms do not resolve, or if symptoms recur on reintroduction of gefitinib, permanent discontinuation should be considered.

In the use of gefitinib and radiation in paediatric patients, with newly diagnosed brain stem glioma or incompletely resected supratentorial malignant glioma, Central Nervous System (CNS) haemorrhages were reported. A further case of CNS haemorrhage has been reported in a child with an ependymoma with gefitinib alone. An increased risk of cerebral haemorrhage in adult patients with NSCLC receiving gefitinib has not been established.

Gastrointestinal perforation has been reported in patients taking gefitinib. In most cases this is associated with other known risk factors, including concomitant medications such as steroids or NSAIDs, underlying history of GI ulceration, age, smoking or bowel metastases at sites of perforation.

Excipients

Gefitinib Sandoz contains Lactose monohydrate. Patients with rare hereditary problems of galactose intolerance, total lactase deficiency or glucose-galactose malabsorption should not take this medicine.

4.5 Interaction with other medicinal products and other forms of interaction

The metabolism of gefitinib is via the cytochrome P450 isoenzyme CYP3A4 (predominantly) and via CYP2D6.

Active substances that may increase gefitinib plasma concentrations

Gefitinib is a substrate of p-glycoprotein (Pgp).

Substances that inhibit CYP3A4 may decrease the clearance of gefitinib. Concomitant administration with potent inhibitors of CYP3A4 activity (e.g. ketoconazole, posaconazole, voriconazole, protease inhibitors, clarithromycin, telithromycin) may increase gefitinib plasma concentrations. The increase may be clinically relevant since adverse reactions are related to dose and exposure. The increase might be higher in individual patients with CYP2D6 poor metaboliser genotype. Pre-treatment with itraconazole (a potent CYP3A4 inhibitor) resulted in an 80% increase in the mean AUC of gefitinib. In situations of concomitant treatment with potent inhibitors of CYP3A4 the patient should be closely monitored for gefitinib adverse reactions.

If concomitant treatment with a potent CYP2D6 inhibitor is initiated, the patient should be closely monitored for adverse reactions.

Active substances that may reduce gefitinib plasma concentrations

Substances that are inducers of CYP3A4 activity may increase metabolism and decrease gefitinib plasma concentrations and thereby reduce the efficacy of gefitinib. Concomitant medicinal products that induce CYP3A4 (e.g. phenytoin, carbamazepine, rifampicin, barbiturates or St John's wort, *Hypericum perforatum*) should be avoided. Pre-treatment with rifampicin (a potent CYP3A4 inducer) in healthy volunteers reduced mean gefitinib AUC by 83%.

Substances that cause significant sustained elevation in gastric pH may reduce gefitinib plasma concentrations and thereby reduce the efficacy of gefitinib. High doses of short-acting antacids may have a similar effect if taken regularly close in time to administration of gefitinib. Concomitant administration of gefitinib with ranitidine at a dose that caused sustained elevations in gastric pH ≥ 5 resulted in a reduced mean gefitinib AUC by 47%.

Active substances that may have their plasma concentrations altered by gefitinib

Gefitinib has limited potential to inhibit CYP2D6. Gefitinib was co-administered with metoprolol (a CYP2D6 substrate). This resulted in a 35% increase in exposure to metoprolol. Such an increase might potentially be relevant for CYP2D6 substrates with narrow therapeutic index. When the use of CYP2D6 substrates are considered in combination with gefitinib, a dose modification of the CYP2D6 substrate should be considered especially for products with a narrow therapeutic window.

Other potential interactions

INR elevations and/or bleeding events have been reported in some patients concomitantly taking warfarin.

4.6 Fertility, pregnancy and lactation

Women of childbearing potential

Women of childbearing potential must be advised not to get pregnant during therapy.

Pregnancy

There are no data from the use of gefitinib in pregnant women. The potential risk for humans is unknown. Gefitinib Sandoz should not be used during pregnancy unless clearly necessary.

Breast-feeding

It is not known whether gefitinib is secreted in human milk. Gefitinib is contraindicated during breast-feeding and therefore breast-feeding must be discontinued while receiving gefitinib therapy.

4.7 Effects on ability to drive and use machines

During treatment with gefitinib, asthenia has been reported. Therefore, patients who experience this symptom should be cautious when driving or using machines.

4.8 Undesirable effects

Summary of the safety profile

The most frequently reported adverse drug reactions (ADRs), occurring in more than are diarrhoea and skin reactions (including rash, acne, dry skin and pruritus). ADRs usually occur within the first month of therapy and are generally reversible.

Interstitial lung disease (ILD) has also occurred in patients, often severe (CTC grade 3-4).

Tabulated list of adverse reactions

Frequencies of occurrence of undesirable effects are defined as: very common; common; uncommon; rare; very rare, not known.

Table 1 Adverse reactions

Adverse reactions by system organ class and frequency		
Metabolism and nutrition disorders	<i>Very common</i>	Anorexia mild or moderate (CTC grade 1 or 2).
Eye disorders	<i>Common</i>	Conjunctivitis, blepharitis, and dry eye*, mainly mild (CTC grade 1).
	<i>Uncommon</i>	Corneal erosion, reversible and sometimes in association with aberrant eyelash growth.
		Keratitis
Vascular disorders	<i>Common</i>	Haemorrhage, such as epistaxis and haematuria.
Respiratory, thoracic and mediastinal disorders	<i>Common</i>	Interstitial lung disease, often severe (CTC grade 3-4). Cases with fatal outcomes have been reported.
Gastrointestinal disorders	<i>Very common</i>	Diarrhoea, mainly mild or moderate (CTC grade 1 or 2).
		Vomiting, mainly mild or moderate (CTC grade 1 or 2).
		Nausea, mainly mild (CTC grade 1).
		Stomatitis, predominantly mild in nature (CTC grade 1).
	<i>Common</i>	Dehydration, secondary to diarrhoea, nausea, vomiting or anorexia.
		Dry mouth*, predominantly mild (CTC grade 1).
	<i>Uncommon</i>	Pancreatitis; Gastrointestinal perforation
Hepatobiliary disorders	<i>Very common</i>	Elevations in alanine aminotransferase, mainly mild to moderate.
	<i>Common</i>	Elevations in aspartate aminotransferase, mainly mild to moderate.
		Elevations in total bilirubin, mainly mild to moderate.
	<i>Uncommon</i>	Hepatitis**
Skin and subcutaneous tissue disorders	<i>Very common</i>	Skin reactions, mainly a mild or moderate (CTC grade 1 or 2)
	<i>Common</i>	Nail disorder
		Alopecia
		Allergic reactions, including angioedema and urticaria

	<i>Uncommon</i>	Palmar-plantar erythrodysesthesia
	<i>Rare</i>	Bullous conditions including Toxic epidermal necrolysis, Stevens Johnson syndrome and erythema multiforme
		Cutaneous vasculitis
Renal and urinary disorders	<i>Common</i>	Asymptomatic laboratory elevations in blood creatinine
		Proteinuria
		Cystitis
General disorders and administration site conditions	<i>Rare</i>	Haemorrhagic cystitis
	<i>Very common</i>	Asthenia, predominantly mild (CTC grade 1).
	<i>Common</i>	Pyrexia

The frequency of adverse drug reactions relating to abnormal laboratory values is based on patients with a change from baseline of 2 or more CTC grades in the relevant laboratory parameters.

*This adverse reaction can occur in association with other dry conditions (mainly skin reactions) seen with gefitinib.

**This includes isolated reports of hepatic failure which in some cases led to fatal outcomes.

4.9 Overdose

There is no specific treatment in the event of overdose of gefitinib. In patients who were treated with daily doses of up to 1000 mg, an increase of frequency and severity of some adverse reactions was observed, mainly diarrhoea and skin rash. Adverse reactions associated with overdose should be treated symptomatically; in particular severe diarrhoea should be managed as clinically indicated.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: antineoplastic agents, protein kinase inhibitors; ATC code: L01EB01

Mechanism of action and pharmacodynamic effects

The epidermal growth factor (EGF) and its receptor (EGFR [HER1; ErbB1]) have been identified as key drivers in the process of cell growth and proliferation for normal and cancer cells. EGFR activating mutation within a cancer cell is an important factor in promotion of tumour cell growth, blocking of apoptosis, increasing the production of angiogenic factors and facilitating the processes of metastasis.

Gefitinib is a selective small molecule inhibitor of the epidermal growth factor receptor tyrosine kinase and is an effective treatment for patients with tumours with activating mutations of the EGFR tyrosine kinase domain regardless of line of therapy. No clinically relevant activity has been shown in patients with known EGFR mutation-negative tumours.

Resistance

Most NSCLC tumours with sensitising EGFR kinase mutations eventually develop resistance to

Gefitinib Sandoz treatment, with a median time to disease progression of 1 year. Other potential mechanisms of resistance that have been reported following treatment with EGFR signal blocking agents include: bypass signalling such as HER2 and MET gene amplification and PIK3CA mutations. Phenotypic switch to small cell lung cancer has also been reported.

Circulating Tumour DNA (ctDNA)

In EGFR mutation-positive patients, significantly more gefitinib-treated patients experienced an improvement in quality of life and lung cancer symptoms vs carboplatin/paclitaxel.

Gefitinib demonstrated superior PFS (progression free survival), ORR (objective response rate), QoL (quality of life) and symptom relief with no significant difference in overall survival compared to carboplatin/paclitaxel in previously untreated patients, with locally advanced or metastatic NSCLC, whose tumours harboured activating mutations of

the EGFR tyrosine kinase.

Pretreated patients

In the overall population, no statistically significant difference between gefitinib and docetaxel (75 mg/m²) was observed for overall survival, progression free survival and objective response rates in patients with locally advanced or metastatic NSCLC who had previously received platinum-based chemotherapy.

5.2 Pharmacokinetic properties

Absorption

Following oral administration of gefitinib, absorption is moderately slow and peak plasma concentrations of gefitinib typically occur at 3 to 7 hours after administration. Mean absolute bioavailability is 59% in cancer patients. Exposure to gefitinib is not significantly altered by food. In a trial in healthy volunteers where gastric pH was maintained above pH 5, gefitinib exposure was reduced by 47%, likely due to impaired solubility of gefitinib in the stomach.

Distribution

Gefitinib has a mean steady-state volume of distribution of 1400 l indicating extensive distribution into tissue. Plasma protein binding is approximately 90%. Gefitinib binds to serum albumin and alpha 1-acid glycoprotein.

Gefitinib is a substrate for the membrane transport protein Pg-p.

Biotransformation

CYP3A4 and CYP2D6 are the major P450 isozyme involved in the oxidative metabolism of gefitinib.

Gefitinib has limited potential to inhibit CYP2D6. Gefitinib shows no enzyme induction effects in animal studies and no significant inhibition (*in vitro*) of any other cytochrome P450 enzyme.

Gefitinib is extensively metabolised in humans. Five metabolites have been fully identified in excreta and 8 metabolites in plasma. The major metabolite identified was O-desmethyl gefitinib, which is 14- fold less potent than gefitinib at inhibiting EGFR stimulated cell growth and has no inhibitory effect on tumour cell growth in mice. It is therefore considered unlikely that it contributes to the clinical activity of gefitinib.

The formation of O-desmethyl gefitinib has been shown, *in vitro*, to be via CYP2D6. In poor metabolisers no measurable levels of O-desmethyl gefitinib were produced. The levels of exposure to gefitinib achieved in both the extensive and the poor metaboliser groups were wide and overlapping but the mean exposure to gefitinib was 2-fold higher in the poor metaboliser group. The higher average exposures that could be achieved by individuals with no active CYP2D6 may be clinically relevant since adverse effects are related to dose and exposure.

Elimination

Gefitinib is excreted mainly as metabolites via the faeces, with renal elimination of gefitinib and metabolites accounting for less than 4% of the administered dose.

Gefitinib total plasma clearance is approximately 500 ml/min and the mean terminal half-life is 41 hours in cancer patients. Administration of gefitinib once daily results in 2- to 8-fold accumulation, with steady state exposures achieved after 7 to 10 doses. At steady state, circulating plasma concentrations are typically maintained within a 2- to 3-fold range over the 24-hour dosing interval.

Special populations

No relationships were identified between predicted steady-state trough concentration and patient age, body weight, gender, ethnicity or creatinine clearance (above 20 ml/min).

Hepatic impairment

There was an increase in exposure in patients with mild, moderate or severe hepatic impairment due to cirrhosis (according to Child-Pugh classification) who received single dose gefitinib 250mg. This increase in exposure may be of clinical relevance since adverse experiences are related to dose and exposure to gefitinib.

Following daily administration of 250 mg gefitinib, time to steady-state, total plasma clearance (C_{max}SS) and

steady-state exposure (AUC_{24SS}) were similar for the groups with normal and moderately impaired hepatic function.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Tablet core

Lactose monohydrate
Microcrystalline cellulose (E460)
Croscarmellose sodium (E468)
Povidone K30 (E1201)
Magnesium stearate (E470b)
Sodium Laurilsulfate

Tablet coating

Poly (Vinyl Alcohol) (E1203)
Macrogol 3350 (E1521)
Talc (E553b)
Titanium Dioxide (E171)
Iron oxide red (E172)
Iron oxide yellow (E172)

6.2 Shelf life

4 years

6.3 Special precautions for storage

Do not store above 30°C.

6.4 Nature and contents of container

Gefitinib Sandoz film-coated tablets are packed in Aluminium-OPA/Alu/PVC blisters. The blisters are packed into carton boxes along with the package insert.

Pack size of 30's, 60's and 100's film-coated tablets. Not all pack sizes may be marketed.

6.5 Special precautions for disposal

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

7. PRODUCT REGISTRATION HOLDER

Sandoz Products Malaysia Sdn. Bhd.
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47400 Petaling Jaya, Selangor, Malaysia

8. DATE OF REVISION OF THE TEXT

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